

Linear Algebra HW 3

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Question 1

The possibilities for i, j are 2, 6 and for k, l they are 4, 6

An odd permutation is one which has an odd number of inversions.

We only need to calculate the sign of one of the possible permutations, as then we know if we switch one entry the sign switches.

For the permutation, $i = 2, \quad j = 6, \quad k = 4, \quad l = 6$

$$\begin{bmatrix} 3 & 2 & 6 & 1 & 4 & 7 & 5 & 8 \\ 3 & 6 & 8 & 7 & 1 & 5 & 2 & 4 \end{bmatrix}$$

We then count the number of inversions, and find that there are 7 on top and 17 on the bottom, therefore the inversion is even.

Then if we switch to either just the i, j entries or just the k, l entries we will have an odd inversion, as when we swap both, two switches of -1 just leave the permutation's sign unchanged.

Thus the values of i, j, k, l which represent an odd permutation are 2, 6, 6, 4 and 6, 2, 4, 6

Question 2

We know that if a matrix is not invertible then the determinant is zero.

Thus we are looking for the values of k which lead to a zero determinant

$$A = \begin{bmatrix} 2 & k & k \\ k & k & k \\ 8 & 7 & k \end{bmatrix}$$

Then

$$\begin{aligned} \det(A) &= 2 \det \begin{bmatrix} k & k \\ 7 & k \end{bmatrix} - k \det \begin{bmatrix} k & k \\ 8 & k \end{bmatrix} + k \det \begin{bmatrix} k & k \\ 8 & 7 \end{bmatrix} \\ &= 2(k^2 - 7k) - k(k^2 - 8k) + k(7k - 8k) \\ &= -k^3 + 9k^2 - 14k \end{aligned}$$

$$k^3 - 9k^2 + 14k = 0$$

$$\Rightarrow k = 0, \quad k = 2, \quad k = 7$$

Question 3

We need the determinant of

$$A = \begin{bmatrix} 0 & 9 & 3 & 9 \\ 2 & 2 & 0 & 2 \\ 3 & 0 & 1 & 3 \\ 5 & -1 & 8 & 7 \end{bmatrix}.$$

Swap R_1 and R_2 , this changes the determinant of A by a sign

$$\begin{bmatrix} 2 & 2 & 0 & 2 \\ 0 & 9 & 3 & 9 \\ 3 & 0 & 1 & 3 \\ 5 & -1 & 8 & 7 \end{bmatrix}.$$

$R_1 \times \frac{1}{2}$, this scales the determinant by $\frac{1}{2}$

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 9 & 3 & 9 \\ 3 & 0 & 1 & 3 \\ 5 & -1 & 8 & 7 \end{bmatrix}.$$

$R_3 \rightarrow R_3 - 3R_1$, $R_4 \rightarrow R_4 - 5R_1$. This doesn't affect the determinant

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 9 & 3 & 9 \\ 0 & -3 & 1 & 0 \\ 0 & -6 & 8 & 2 \end{bmatrix}.$$

$R_3 \rightarrow R_3 + \frac{1}{3}R_2$, $R_4 \rightarrow R_4 + \frac{2}{3}R_2$. Similarly this doesn't affect the determinant

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 9 & 3 & 9 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 10 & 8 \end{bmatrix}.$$

$R_4 \rightarrow R_4 - 5R_3$.

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 9 & 3 & 9 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & -7 \end{bmatrix}.$$

The determinant of a matrix in upper triangular form is simply the product of the elements along the main diagonal, but don't forget the changes as a result of the row operations

$$\det(A) = (1)(9)(2)(-7) \times (2)(-1) = 252.$$

Question 4

We need the determinant of

$$B = \begin{bmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 4 & 5 & 1 & 2 \\ 1 & 0 & -2 & 2 & 3 \\ 3 & -1 & 1 & 0 & 4 \\ 2 & 2 & 3 & 1 & 1 \end{bmatrix}.$$

$R_1 \times \frac{1}{2}$, this scales the determinant by $\frac{1}{2}$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 4 & 5 & 1 & 2 \\ 1 & 0 & -2 & 2 & 3 \\ 3 & -1 & 1 & 0 & 4 \\ 2 & 2 & 3 & 1 & 1 \end{bmatrix}.$$

$R_3 \rightarrow R_3 - 1 \cdot R_1$, $R_4 \rightarrow R_4 - 3 \cdot R_1$, $R_5 \rightarrow R_5 - 2 \cdot R_1$, This doesn't effect the determinant

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 4 & 5 & 1 & 2 \\ 0 & -\frac{1}{2} & -2 & \frac{1}{2} & \frac{7}{2} \\ 0 & -\frac{5}{2} & 1 & -\frac{9}{2} & \frac{11}{2} \\ 0 & 1 & 3 & -2 & 2 \end{bmatrix}.$$

$R_2 \times \frac{1}{4}$ This scales the determinant by $\frac{1}{4}$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{5}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & -\frac{1}{2} & -2 & \frac{1}{2} & \frac{7}{2} \\ 0 & -\frac{5}{2} & 1 & -\frac{9}{2} & \frac{11}{2} \\ 0 & 1 & 3 & -2 & 2 \end{bmatrix}.$$

$$R_3 \rightarrow R_3 + \frac{1}{2}R_2, \quad R_4 \rightarrow R_4 + \frac{5}{2}R_2, \quad R_5 \rightarrow R_5 - 1 \cdot R_2,$$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{5}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & -\frac{11}{8} & \frac{5}{8} & \frac{15}{4} \\ 0 & 0 & \frac{33}{8} & -\frac{31}{8} & \frac{27}{4} \\ 0 & 0 & \frac{7}{4} & -\frac{9}{4} & \frac{3}{2} \end{bmatrix}.$$

$$R_4 \rightarrow R_4 + 3R_3, \quad R_5 \rightarrow R_5 + \frac{14}{11}R_3,$$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{5}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & -\frac{11}{8} & \frac{5}{8} & \frac{15}{4} \\ 0 & 0 & 0 & -2 & 18 \\ 0 & 0 & 0 & -\frac{16}{11} & \frac{69}{11} \end{bmatrix}.$$

$$R_5 \rightarrow R_5 - \frac{8}{11}R_4,$$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{5}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & -\frac{11}{8} & \frac{5}{8} & \frac{15}{4} \\ 0 & 0 & 0 & -2 & 18 \\ 0 & 0 & 0 & 0 & -\frac{75}{11} \end{bmatrix}.$$

The determinant of a matrix in upper triangular form is simply the product of the elements along the main diagonal, but don't forget the changes as a result of the row operations

$$\det(B) = (1)(1)\left(-\frac{11}{8}\right)(-2)\left(-\frac{75}{11}\right) \times (2)(4) = -150$$

Question 5

$$C^{-1}(A + X)B^{-1} = I_n$$

Since we know $C^{-1}C = I_n$, one can see that

$$(A + X)B^{-1} = C$$

$$\Rightarrow A + X = CB$$

$$\Rightarrow X = CB - A$$

Question 6

This statement is false and can be easily proven with a simple counterexample

Consider the 2×4 augmented matrix of a linear system of two equations in three variables

$$\begin{bmatrix} 2 & 4 & 6 & 1 \\ 4 & 8 & 12 & 25 \end{bmatrix}$$

Then when we do the row operation $r_2 - 2r_1$, we get

$$\begin{bmatrix} 2 & 4 & 6 & 1 \\ 0 & 0 & 0 & 23 \end{bmatrix}$$

There is an obvious inconsistency of $0 = 23$, which implies the system of linear equations has no solutions. Thus the original statement is false.

Question 7

Since the linear system, with $\vec{b} = \vec{0}$ has an infinite number of solutions this means that in A there is not going to be a pivot variable in each column, thus there are going to be at least three zero rows when we row reduce A .

Then in the second linear system when we use row reduction and get at least three rows of zero in A , where at least one of them could be equal to a non-zero constant, which makes the system inconsistent and have no solutions.

Or the other possibility is during the row reduction we get an additional zero in $\vec{b}$ in which case there are no inconsistencies, still not a pivot variable in each column and thus an infinite number of solutions.

Thus the two possibilities are zero solutions and infinitely many solutions.

Question 8

We know that the following properties of the determinant of a matrix are true

- $\det(AB) = (\det(A))(\det(B))$
- $\det(A^T) = \det(A)$

Then

1. $\det(ABAC) = (\det(AB)) (\det(AC)) = \det(A)^2 (\det(B)) (\det(C)) = -200$
2. $\det(C^4) = \det(CCCC) = \det(C)^4 = 16$
3. $\det(B^T BC) = (\det(B^T)) (\det(B)) (\det(C)) = \det(B)^2 (\det(C)) = -32$

Question 9

There are a few facts which are relevant to this question, when you perform the elementary row operation of

- replacement, the determinant of the new matrix remains unchanged
- interchanging, the determinant of the new matrix is -1 times the previous determinant
- scaling, the determinant of the new matrix is also scaled by the same constant

Then the elementary matrices have the following effects on the determinants of the new matrices formed

- E_1 has no effect (replacement)
- E_2 scales the determinant by $-\frac{4}{7}$ (scaling)
- E_3 changes the sign of the determinant (interchanging)
- E_4 has no effect (replacement)

Thus

1. $\det(E_4 E_3) = -1$
2. $\det(E_4 E_1) = 1$
3. $\det(E_2 E_4) = -\frac{4}{7}$
4. $\det(E_1 E_2 E_3 E_4 A) = (1)(-\frac{4}{7})(-1)(1)(5) = \frac{20}{7}$